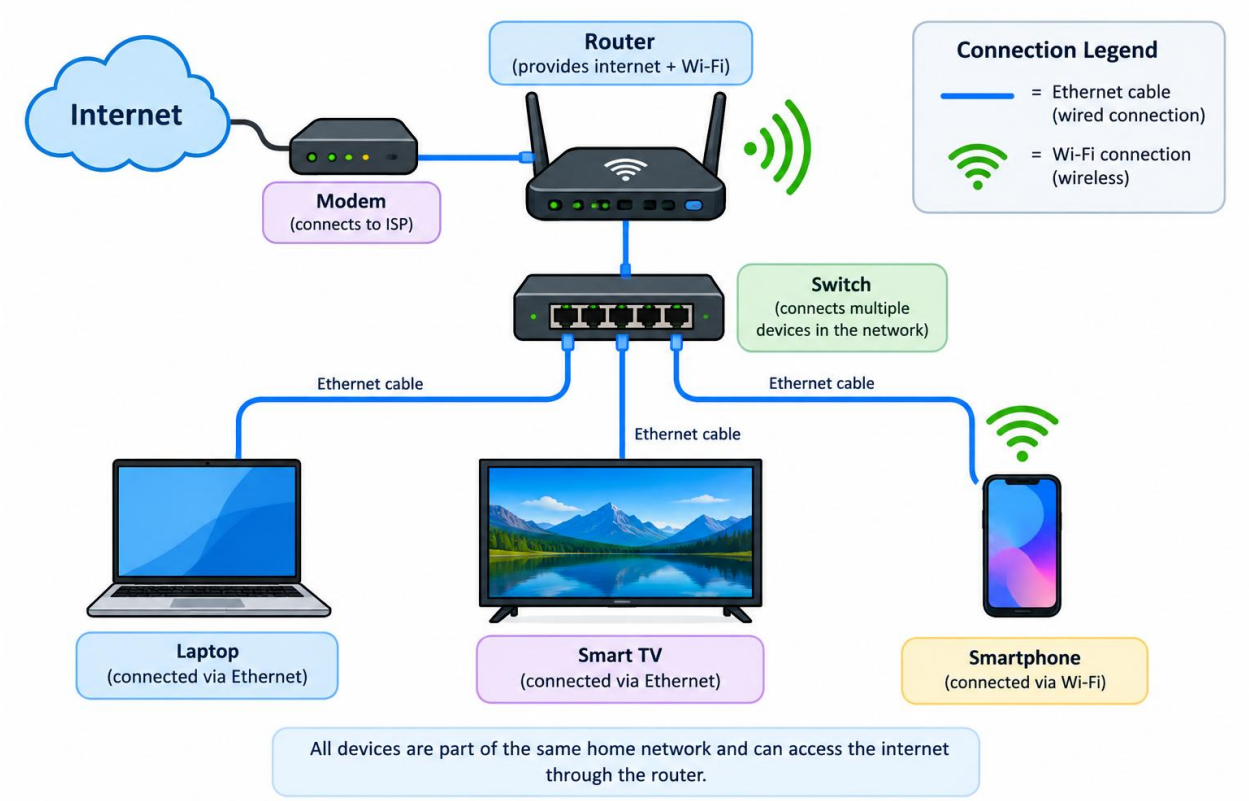
1.Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Ans.



2.Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Ans.

Network Device	Main Work	Real-World Use Case
1. Router	Connects different networks and provides internet access.	We use a Wi-Fi router to watch videos on YouTube or use Instagram on our phones.
2. Switch	Connects multiple devices in a local network (LAN).	In a computer lab, a switch connects computers so students can share files and access network resources.
3. Hub	Connects multiple devices and sends received data to all connected devices.	In an older computer lab, a hub can connect several computers for basic LAN communication.
4. Access Point (AP)	Provides Wi-Fi access to a network.	In a college, an access point allows students to connect their phones and laptops to Wi-Fi to use WhatsApp or Google Classroom.

3.Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Ans. Scenario: Ordering Food on Zomato

When I order food on Zomato using my phone, the data travels like this:

1. **Access Point: My phone connects to Wi-Fi through the access point, which allows my phone to send data wirelessly.**
2. **Router: The router sends my order data from my home network to the internet.**

3. **Switch: Switches help transfer data between devices in the local network and within the internet service provider's network.**
4. **Internet and Zomato Server: My order reaches Zomato's server, which processes the request and sends the confirmation back to my phone.**

4.Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Think about devices like switches, powerline adapters, or WiFi extenders.

Ans. You can use these two devices to connect your gaming PC to the internet:

1. Powerline Adapter

- **Connect one adapter to the router using an Ethernet cable.**
- **Plug another adapter near your gaming PC and connect it to the PC.**
- **It sends internet data through your home's electrical wiring.**
- **Benefit: It can provide a stable connection for online gaming.**

2. Wi-Fi Extender

- **Place the extender near your gaming PC where it gets a good Wi-Fi signal.**
- **Connect the extender to your Wi-Fi router.**
- **Use an Ethernet cable to connect the extender to your gaming PC.**
- **Benefit: It helps your PC get internet without using a long Ethernet cable.**

Conclusion: Both devices can connect your gaming PC to the internet. A powerline adapter may be better for gaming if it provides a stable connection in your home.